Skill Check 1: Basics of R

Exam Agreement

This Skill Check is an individual assessment and you should not receive or offer help on it from any other human. Cell phones are to be handed in to the instructor and all other digital devices must be stored in bags underneath the bench of in the lab cubbies.

You may use any **physical** resource to complete the work. This includes:

- Any notes, code, slides, papers, or previous feedback from the instructor as long as they are on paper.
- Any books that you have with you.
- Any scholarly works such as papers that you have with you.

You may NOT use:

- Help from any other student or person. **This is an individual assessment.**
- Any digital resource that does not exist as a physical copy present in class.
- The use of generative artificial intelligence (e.g., ChatGPT).
- Help from homework websites such as Course Hero or Chegg.

By signing, you agree that you have neither given nor received unauthorized aid on this examination.

Printed Name: ____

Chapman ID: ____

Signed: ____

Date: ____________________

Include this signed page as the first page of your submitted work.

Skill Check Instructions

You must answer every question on the exam to the best of your ability. Include all appropriate code, syntax, and functions that would lead to the code to run successfully.

Question 1: How Computers Work

What is a vectorized calculation? Describe it briefly in words, then provide an example of a vectorized calculation in R.

Question 2: Debugging and Troubleshooting

The following line of code does not work:

```
my_list < - list("actions" = c("jump", "block", "kick"), sport = c("soccer",
"baseball") matrix(c(T,T, T,F, T,F)))
```

It produces the following error:

```
Error: unexpected symbol in:
"my_list < - list("actions" = c("jump", "block", "kick"), sport = c("soccer",
                                                            "baseball") matrix"
```

Rewrite the line of code that corrects the error and runs successfully below.

Question 3: Data Frames

Write a line of code that calculates the mean uptake of CO_2 for a plant grown in Mississippi receiving the chilled treatment in the `CO2` data set.

```
CO2                84 obs. of 5 variables
    $ Plant    : Ord.factor w/ 12 levels "Qn1"<"Qn2"<"Qn3"<.…
    $ Type     : Factor w/ 2 levels "Quebec","Mississippi": …
    $ Treatment: Factor w/ 2 levels "nonchilled","chilled": …
    $ conc     : num  95 175 250 350 500 675 1000 95 175 250…
    $ uptake   : num  16 30.4 34.8 37.2 35.3 39.2 39.7 13.6 …
    - attr(*, "formula")=Class 'formula'  language uptake ~ …
     .. ..- attr(*, ".Environment")=<environment: R_EmptyEnv…
    - attr(*, "outer")=Class 'formula'  language ~Treatment …
     .. ..- attr(*, ".Environment")=<environment: R_EmptyEnv…
```

Question 4: Matrices and Arrays

Write a line of code that creates a 6-dimensional array with any number of dimensions using `runif()` to generate its data.

Question 5: Lists

I have created a list stored with the name `my_list` with the following code:

```
my_list <- list("blep" = c("a", "b", "c", "d", "e"),
                "freyja" = array(runif(300), dim = c(3, 10, 10)),
                "kessel" = data.frame("x" = c(1.1, 1.2, 1.3),
                                      "y" = c(2.3, 5.6, 6.2)),
                 1L)
```

a. How many members does this list contain?

b. What data class is each member?

c. How would you reference the letter “c” in `my_list`?

Question 6: Class Coercion

To count the number of `NA`'s in the data frame `df`, you can use `sum(is.na(df))`. Explain how this works, including any class coercions that might happen.

Question 7: Data Classes and Special Values

The following is not a valid vector in R. Describe why this is the case.

```
vec <- c(4, 8, NULL)
```

Question 8: Factors

Write a line of code that creates an ordered factor that rates your opinion of the following sports: basketball, soccer, water polo, volleyball, football, cross country. Your rating system should have at least three levels.

Question 9: Vectors

What is an example of a function that can be used to produce a vector? Describe what this function does and/or produces. Give a working example.